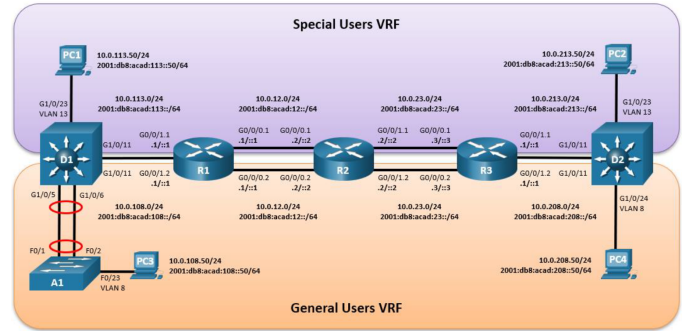


ENCOR Skills Assessment (Scenario 2)

1st Camilo Andres Gutierrez Rodriguez
Facultad de Ingeniería, Diseño e Innovación
Institución Universitaria Politécnico Gran Colombiano
Bogotá D.C, Colombia
candresgutierrez1@poligran.edu.co

Abstract—This project describes the deployment of a multi-VRF network designed to support isolated user environments on a shared physical infrastructure. Two VRF instances were established, General Users and Special Users, in order to provide independent routing domains with full end-to-end reachability inside each VRF and no inter-VRF communication. Technologies such as VRF-Lite, VLAN segmentation, router-on-a-stick, and static routing for IPv4 and IPv6 are used for the implementation utilizing GNS3. The development of this project shows how VRFs can be implemented in enterprise networks to achieve logical division, higher security, and more appropriate traffic management.

Index Terms—VRF, VLAN segmentation, router-on-a-stick, routing, security, traffic management, GNS3.



Teniendo en cuenta lo anterior, la topología implementada para el desarrollo del proyecto es la siguiente:

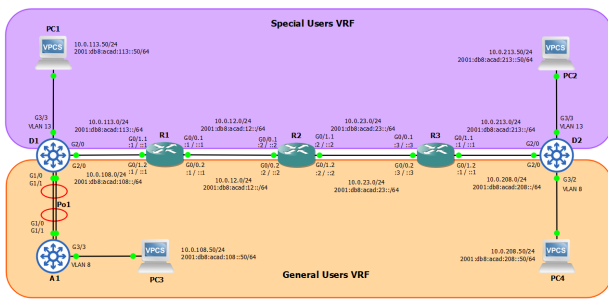


Fig. 4. Topología implementada

III. CONFIGURE BASIC SETTINGS FOR EACH DEVICE.

Para esta sección, se configuraron los parámetros básicos en los dispositivos, en los cuales se encuentra: Nombre, banner, desactivar búsqueda de domino, habilitar enrutamiento IPv6, configuraciones de login en consola, y en el caso de los switches la creación de las VLAN (8 y 13)

IV. PART 2: CONFIGURE VRF AND STATIC ROUTING

Para el desarrollo de esta sección, se procede con la configuración de VRF-Lite en los tres routers, y adicionalmente de las rutas estáticas adecuadas para lograr el alcance punto a punto (end-to-end), verificando con pruebas de conectividad. Para ello se desarrollaron diferentes tareas de configuración, con el fin de que el proceso sea adecuado.

A. On R1, R2, and R3, configure VRF-Lite VRFs as shown in the topology diagram.

Esta tarea se desarrolló creando las VRFs, General Users y Special Users, en los tres routers, soportando IPv4 e IPv6. Las configuraciones realizadas son las siguientes, se incluye comando de verificación:

```
R2(config)#vrf definition General-Users
R2(config-vrf)# !
R2(config-vrf)# address-family ipv4
R2(config-vrf-af)# exit-address-family
R2(config-vrf)# !
R2(config-vrf)# address-family ipv6
R2(config-vrf-af)# exit-address-family
R2(config-vrf)#vrf definition Special-Users
R2(config-vrf)# !
R2(config-vrf)# address-family ipv4
R2(config-vrf-af)# exit-address-family
R2(config-vrf)# !
R2(config-vrf)# address-family ipv6
R2(config-vrf-af)# exit-address-family
R2(config-vrf)#ex
R2(config)#do sh vrf bri
```

Name	Default RD	Protocols	Interfaces
General-Users	<not set>	ipv4,ipv6	
Special-Users	<not set>	ipv4,ipv6	

Fig. 5. Creación VRFs

Es necesario aclarar, que en caso de que no sea posible crear las vrf ni aplicarlas posteriormente, esto se debe a que probablemente no está habilitado "ip cef", lo cual impide la configuración de las VRFs, para habilitarlo, dentro del modo de configuración global debe habilitarse "ip cef" e "ipv6 cef", con esto se puede proceder a la configuración de VRFs correspondientes.

B. On R1, R2, and R3, configure IPv4 and IPv6 interfaces on each VRF as detailed in the addressing table above.

Para realizar estas configuraciones adecuadamente, primero que todo, se tiene que realizar el encapsulamiento para VLAN a configurar en cada subinterfaz (Router-On-A-Stick), luego se procede a aplicar la VRF a las subinterfaces correspondientes siguiendo el diseño en la topología, y una vez se haya realizado esto, se procede a realizar el direccionamiento y a encender la interfaz. Teniendo en cuenta esto, las VLAN para cada VRF corresponden a:

- VRF Special Users - VLAN 13 - Sub-interfaz 1 (Gix/x.1)
- VRF General Users - VLAN 8 - Sub-interfaz 2 (Gix/x.2)

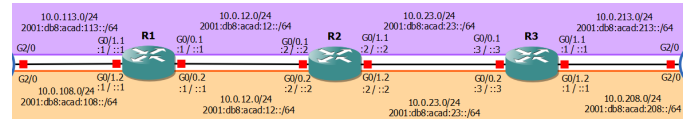


Fig. 6. Routers a configurar

A continuación, se detallan las configuraciones realizadas para cada uno de los routers en la topología

1) Router - R1:

```
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/0.1
encapsulation dot1Q 13
vrf forwarding Special-Users
ip address 10.0.12.1 255.255.255.0
ipv6 address FE80::1:1 link-local
ipv6 address 2001:DB8:ACAD:12::1/64
!
interface GigabitEthernet0/0.2
encapsulation dot1Q 8
vrf forwarding General-Users
ip address 10.0.12.1 255.255.255.0
ipv6 address FE80::1:2 link-local
ipv6 address 2001:DB8:ACAD:12::1/64
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/1.1
encapsulation dot1Q 13
vrf forwarding Special-Users
ip address 10.0.113.1 255.255.255.0
ipv6 address FE80::1:3 link-local
ipv6 address 2001:DB8:ACAD:113::1/64
!
interface GigabitEthernet0/1.2
encapsulation dot1Q 8
vrf forwarding General-Users
ip address 10.0.108.1 255.255.255.0
ipv6 address FE80::1:4 link-local
ipv6 address 2001:DB8:ACAD:108::1/64
```

Fig. 7. Configuración de R1: Router-on-a-Stick, VRFs y direccionamiento

2) Router - R2:

```
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/0.1
encapsulation dot1Q 13
vrf forwarding Special-Users
ip address 10.0.12.2 255.255.255.0
ipv6 address FE80::2:1 link-local
ipv6 address 2001:DB8:ACAD:12::2/64
!
interface GigabitEthernet0/0.2
encapsulation dot1Q 8
vrf forwarding General-Users
ip address 10.0.12.2 255.255.255.0
ipv6 address FE80::2:2 link-local
ipv6 address 2001:DB8:ACAD:12::2/64
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/1.1
encapsulation dot1Q 13
vrf forwarding Special-Users
ip address 10.0.23.2 255.255.255.0
ipv6 address FE80::2:3 link-local
ipv6 address 2001:DB8:ACAD:23::2/64
!
interface GigabitEthernet0/1.2
encapsulation dot1Q 8
vrf forwarding General-Users
ip address 10.0.23.2 255.255.255.0
ipv6 address FE80::2:4 link-local
ipv6 address 2001:DB8:ACAD:23::2/64
```

Fig. 8. Configuración de R2: Router-on-a-Stick, VRFs y direccionamiento

3) Router - R3:

```
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/0.1
encapsulation dot1Q 13
vrf forwarding Special-Users
ip address 10.0.23.3 255.255.255.0
ipv6 address FE80::3:1 link-local
ipv6 address 2001:DB8:ACAD:23::3/64
!
interface GigabitEthernet0/0.2
encapsulation dot1Q 8
vrf forwarding General-Users
ip address 10.0.23.3 255.255.255.0
ipv6 address FE80::3:2 link-local
ipv6 address 2001:DB8:ACAD:23::3/64
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/1.1
encapsulation dot1Q 13
vrf forwarding Special-Users
ip address 10.0.213.1 255.255.255.0
ipv6 address FE80::3:3 link-local
ipv6 address 2001:DB8:ACAD:213::1/64
!
interface GigabitEthernet0/1.2
encapsulation dot1Q 8
vrf forwarding General-Users
ip address 10.0.208.1 255.255.255.0
ipv6 address FE80::3:4 link-local
ipv6 address 2001:DB8:ACAD:208::1/64
!
```

Fig. 9. Configuración de R3: Router-on-a-Stick, VRFs y direccionamiento

C. On R1 and R3, configure default static routes pointing to R2.

D. Verify connectivity in each VRF.

Para el desarrollo de estas secciones, se realizaron las siguientes configuraciones de rutas estáticas totalmente especificadas en R1 y R3, como se solicitó:

1) Router - R1:

```
R1(config)#do sh run | sec ip route
ip route vrf General-Users 0.0.0.0 0.0.0.0 GigabitEthernet0/0.2 10.0.12.2
ip route vrf Special-Users 0.0.0.0 0.0.0.0 GigabitEthernet0/0.1 10.0.12.2
R1(config)#do sh run | sec ipv6 route
ipv6 route vrf Special-Users ::/0 GigabitEthernet0/0.1 2001:DB8:ACAD:12::2
ipv6 route vrf General-Users ::/0 GigabitEthernet0/0.2 2001:DB8:ACAD:12::2
```

Fig. 10. Configuración de rutas estáticas R1

2) Router - R3:

```
R3(config)#do sh run | sec ip route
ip route vrf General-Users 0.0.0.0 0.0.0.0 GigabitEthernet0/0.2 10.0.23.2
ip route vrf Special-Users 0.0.0.0 0.0.0.0 GigabitEthernet0/0.1 10.0.23.2
R3(config)#do sh run | sec ipv6 route
ipv6 route vrf General-Users ::/0 GigabitEthernet0/0.2 2001:DB8:ACAD:23::2
ipv6 route vrf Special-Users ::/0 GigabitEthernet0/0.1 2001:DB8:ACAD:23::2
```

Fig. 11. Configuración de rutas estáticas R3

Siguiendo los pasos solicitados, se logró conectividad de R1 a R3 hasta las interfaces G0/0 en ambas subinterfaces (seleccionadas en un recuadro rojo), sin embargo, no se logra conectividad hacia la siguiente interfaz, la G0/1 (seleccionadas en un recuadro amarillo)

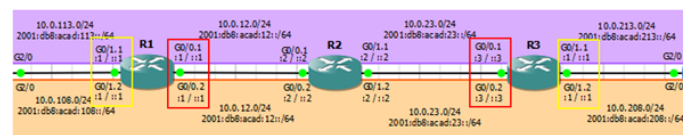


Fig. 12. Alcance de conectividad

Se adjuntan capturas de los resultados al realizar ping y traceroute

3) Pruebas Ping:

```
R1#ping vrf Special-Users 10.0.23.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.23.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/5/13 ms
R1#ping vrf General-Users 10.0.23.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.23.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/5/12 ms
R1#ping vrf General-Users 10.0.213.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.213.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
R1#ping vrf Special-Users 10.0.213.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.213.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
R1#
```

Fig. 13. Prueba de Ping

4) Pruebas Traceroute:

```
R1#trace vrf Special-Users 10.0.213.1
Type escape sequence to abort.
Tracing the route to 10.0.213.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.0.12.2 4 msec 2 msec 14 msec
  2 10.0.12.2 !H !H !H
```

Fig. 14. Prueba de Traza

Se identificó que R2 no estaba aprendiendo las redes mencionadas anteriormente, por tanto, no genera el enrutamiento adecuadamente. Como solución se determinó la implementación de varias rutas estáticas en R2 hacia ambos extremos (R1 y R3) para cada VRF.

5) Router - R2:

```
R2#sh run | sec ip route
ip route vrf General-Users 10.0.108.0 255.255.255.0 GigabitEthernet0/0.2 10.0.12.1
ip route vrf General-Users 10.0.208.0 255.255.255.0 GigabitEthernet0/1.2 10.0.23.3
ip route vrf Special-Users 10.0.113.0 255.255.255.0 GigabitEthernet0/0.1 10.0.12.1
ip route vrf Special-Users 10.0.213.0 255.255.255.0 GigabitEthernet0/1.1 10.0.23.3
R2#sh run | sec ipv6 route
ipv6 route vrf General-Users 2001:DB8:ACAD:108::/64 GigabitEthernet0/0.2 2001:DB8:ACAD:12::1
ipv6 route vrf Special-Users 2001:DB8:ACAD:113::/64 GigabitEthernet0/0.1 2001:DB8:ACAD:12::1
ipv6 route vrf General-Users 2001:DB8:ACAD:208::/64 GigabitEthernet0/1.2 2001:DB8:ACAD:23::3
ipv6 route vrf Special-Users 2001:DB8:ACAD:213::/64 GigabitEthernet0/1.1 2001:DB8:ACAD:23::3
```

Fig. 15. Configuración de rutas estáticas R2

Luego de realizar estas configuraciones, se procedió a verificar los cambios en la tabla de enrutamiento, a continuación, se muestra el antes y después de esta tabla en R2

6) Tabla de Enrutamiento Previa IPv4 - R2:

```
R2(config)#DO SH IP ROUTE vrf General-Users

Routing Table: General-Users
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/24 is directly connected, GigabitEthernet0/0.2
L    10.0.12.2/32 is directly connected, GigabitEthernet0/0.2
C    10.0.23.0/24 is directly connected, GigabitEthernet0/1.2
L    10.0.23.2/32 is directly connected, GigabitEthernet0/1.2
R2(config)#DO SH IP ROUTE vrf Special-Users

Routing Table: Special-Users
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/24 is directly connected, GigabitEthernet0/0.1
L    10.0.12.2/32 is directly connected, GigabitEthernet0/0.1
C    10.0.23.0/24 is directly connected, GigabitEthernet0/1.1
L    10.0.23.2/32 is directly connected, GigabitEthernet0/1.1
```

Fig. 16. Configuración de rutas estáticas R2

7) Tabla de Enrutamiento Previa IPv6 - R2:

```
R2(config)#DO SH IPV6 ROUTE VRF Special-Users
IPv6 Routing Table - Special-Users - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
C    2001:DB8:ACAD:12::/64 [0/0]
       via GigabitEthernet0/0.1, directly connected
L    2001:DB8:ACAD:12::2/128 [0/0]
       via GigabitEthernet0/0.1, receive
C    2001:DB8:ACAD:23::/64 [0/0]
       via GigabitEthernet0/1.1, directly connected
L    2001:DB8:ACAD:23::2/128 [0/0]
       via GigabitEthernet0/1.1, receive
L    FF00::/8 [0/0]
       via Null0, receive
R2(config)#DO SH IPV6 ROUTE VRF General-Users
IPv6 Routing Table - General-Users - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
C    2001:DB8:ACAD:12::/64 [0/0]
       via GigabitEthernet0/0.2, directly connected
L    2001:DB8:ACAD:12::2/128 [0/0]
       via GigabitEthernet0/0.2, receive
C    2001:DB8:ACAD:23::/64 [0/0]
       via GigabitEthernet0/1.2, directly connected
L    2001:DB8:ACAD:23::2/128 [0/0]
       via GigabitEthernet0/1.2, receive
L    FF00::/8 [0/0]
       via Null0, receive
```

Fig. 17. Configuración de rutas estáticas R2

8) Tabla de Enrutamiento Posterior IPv4 - R2:

```
R2(config)#do sh ip route vrf Special-Users

Routing Table: Special-Users
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.12.0/24 is directly connected, GigabitEthernet0/0.1
L    10.0.12.2/32 is directly connected, GigabitEthernet0/0.1
C    10.0.23.0/24 is directly connected, GigabitEthernet0/1.1
L    10.0.23.2/32 is directly connected, GigabitEthernet0/1.1
S    10.0.113.0/24 [1/0] via 10.0.12.1, GigabitEthernet0/0.1
S    10.0.213.0/24 [1/0] via 10.0.23.3, GigabitEthernet0/1.1
R2(config)#do sh ip route vrf General-Users

Routing Table: General-Users
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.12.0/24 is directly connected, GigabitEthernet0/0.2
L    10.0.12.2/32 is directly connected, GigabitEthernet0/0.2
C    10.0.23.0/24 is directly connected, GigabitEthernet0/1.2
L    10.0.23.2/32 is directly connected, GigabitEthernet0/1.2
S    10.0.108.0/24 [1/0] via 10.0.12.1, GigabitEthernet0/0.2
S    10.0.208.0/24 [1/0] via 10.0.23.3, GigabitEthernet0/1.2
```

Fig. 18. Configuración de rutas estáticas R2

9) Tabla de Enrutamiento Posterior IPv6 - R2:

```
R2(config)#DO SH IPV6 ROUTE VRF Special-Users
IPv6 Routing Table - Special-Users - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
ND - ND Default, NDP - ND Prefix, DCE - Destination, NDR - Redirect
RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
lA - LISP away, a - Application
C 2001:DB8:ACAD:12::/64 [0/0]
  via GigabitEthernet0/0.1, directly connected
L 2001:DB8:ACAD:12::2/128 [0/0]
  via GigabitEthernet0/0.1, receive
C 2001:DB8:ACAD:23::/64 [0/0]
  via GigabitEthernet0/1.1, directly connected
L 2001:DB8:ACAD:23::2/128 [0/0]
  via GigabitEthernet0/1.1, receive
S 2001:DB8:ACAD:113::/64 [1/0]
  via 2001:DB8:ACAD:12::1, GigabitEthernet0/0.1
S 2001:DB8:ACAD:213::/64 [1/0]
  via 2001:DB8:ACAD:23::3, GigabitEthernet0/1.1
L FF00::/8 [0/0]

R2(config)#DO SH IPV6 ROUTE VRF General-Users
IPv6 Routing Table - General-Users - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
ND - ND Default, NDP - ND Prefix, DCE - Destination, NDR - Redirect
RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
lA - LISP away, a - Application
C 2001:DB8:ACAD:12::/64 [0/0]
  via GigabitEthernet0/0.2, directly connected
L 2001:DB8:ACAD:12::2/128 [0/0]
  via GigabitEthernet0/0.2, receive
C 2001:DB8:ACAD:23::/64 [0/0]
  via GigabitEthernet0/1.2, directly connected
L 2001:DB8:ACAD:23::2/128 [0/0]
  via GigabitEthernet0/1.2, receive
S 2001:DB8:ACAD:108::/64 [1/0]
  via 2001:DB8:ACAD:12::1, GigabitEthernet0/0.2
S 2001:DB8:ACAD:208::/64 [1/0]
  via 2001:DB8:ACAD:23::3, GigabitEthernet0/1.2
L FF00::/8 [0/0]
  via Null0, receive
```

Fig. 19. Configuración de rutas estáticas R2

Además, se realizaron las pruebas de conectividad nuevamente, las cuales fueron exitosas. Adicionalmente, hasta este punto aún no hay conectividad entre los PCs de extremo a extremo en cada VRF.

10) Pruebas de conectividad posteriores:

```
R1#ping vrf General-Users 10.0.208.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.208.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/6/11 ms
R1#ping vrf General-Users 2001:db8:acad:208::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:208::1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/9 ms
R1#ping vrf Special-Users 10.0.213.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.213.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/6/11 ms
R1#ping vrf Special-Users 2001:db8:acad:213::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:213::1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/8 ms
```

Fig. 20. Pruebas de conectividad posteriores

V. PART 3: CONFIGURE L2 NETWORK

En esta sección se configurarán los switches para brindar la conectividad adecuada y segura a los dispositivos finales. Para

ello, se usarán configuraciones de etherchannel, spanning-tree, puertos troncales y de acceso mediante VLANs (Special Users, General Users). Además, aunque no se solicite, se procede a implementar VLAN Nativa para brindar mayor seguridad al tráfico.

A. On D1, D2, and A1, disable all interfaces.

Para el desarrollo de esta sección se proceden a apagar todas las interfaces y poco a poco, al momento de configurar los demás items, se encienden las interfaces necesarias, de esta forma se brinda seguridad a nivel de capa 2 y físicamente.

```
D2(config)#int ran g0/0-3,g1/0-3,g2/0-3,g3/0-3
D2(config-if-range)#shu
D2(config-if-range)#shutdown
```

Fig. 21. Comando desactivar interfaces en los tres switches

B. On D1 and D2, configure the trunk links to R1 and R3.

Teniendo en cuenta la topología planteada, se procede a troncalizar los puertos G2/0 tanto en el Switch D1 como D2, mediante el uso de la VLAN NATIVA 99 y permitiendo las VLAN 8, 13 y 99 en cuestión, además se asegura el puerto troncal y las negociaciones de DTP mediante el comando switchport nonegotiate, de tal forma que no envíe ni responda tramas DTP.

```
interface GigabitEthernet2/0
switchport trunk allowed vlan 8,13,99
switchport trunk encapsulation dot1q
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
negotiation auto
```

Fig. 22. Configuración Puerto Troncal Switches a Routers

C. On D1 and A1, configure the EtherChannel.

Para realizar estas configuraciones es necesario tener en cuenta los requerimientos. Se solicita usar PAGP para realizar la creación de etherchannel, mediante el Port-channel 1. PAGP posee dos modos de configuración, Auto y Desirable, y se seleccionó el segundo modo respectivamente, para configurar en ambos Switches, de tal forma que se asegura la creación y funcionamiento adecuado de estos enlaces. Para la configuración de etherchannel es necesario identificar los enlaces que se van a unificar en ambos switches. Teniendo en cuenta la topología, son los enlaces G1/0 y G1/1 tanto en D1 como en A1. Primero que todo, hay que realizar las configuraciones básicas para cada interfaz (troncalización con VLAN Nativa, encapsulamiento y desactivación de DTP) luego se configura el channel-group al que van a pertenecer estas interfaces (1) con el modo seleccionado, desirable.

```

interface GigabitEthernet1/0
 switchport trunk allowed vlan 8,13,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 switchport nonegotiate
 negotiation auto
 channel-group 1 mode desirable
interface GigabitEthernet1/1
 switchport trunk allowed vlan 8,13,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 switchport nonegotiate
 negotiation auto
 channel-group 1 mode desirable

```

Fig. 23. Configuración Puertos Troncales Switch A1 a D1 y channel-group

A continuación, en el modo de configuración global es necesario acceder a la interfaz del port-channel 1 y realizar las mismas tareas de troncalización con VLAN Nativa, encapsulamiento y desactivación de DTP

```

interface Port-channel1
 switchport trunk allowed vlan 8,13,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 switchport nonegotiate

```

Fig. 24. Configuración Interfaz Port-Channel 1

Ahora se procede a verificar la creación del channel-group en ambos switches, como se puede evidenciar en la imagen, el enlace se encuentra activo

```

A1#sh etherchannel summary
Flags: D - down P - bundled in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use M - not in use, no aggregation
f - failed to allocate aggregator

M - not in use, minimum links not met
m - not in use, port not aggregated due to minimum links not met
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators: 1

Group Port-channel Protocol Ports
-----
1 Po1(SU) PAgP Gi1/0(P) Gi1/1(P)

```

Fig. 25. Verificación de creación de Channel-group 1

D. On D1, D2, and A1, configure access ports for PC1, PC2, PC3, and PC4.

Para realizar estas configuraciones, es necesario acceder a los puertos que conectan a los PCs, encenderlos y configurar el puerto de acceso y VLAN que le corresponde a cada interfaz. Adicionalmente, para que el puerto encienda de forma

más rápida, y ya que es seguro que el dispositivo que se va a conectar es un PC y no un dispositivo L2 o L3, se procederá a habilitar portfast en las interfaces conectadas a dispositivos finales. Para ello es necesario tener en cuenta el diseño planteado, con el fin de aplicar las VLANs a las interfaces que les corresponden

```

interface GigabitEthernet3/3
 switchport access vlan 13
 switchport mode access
 negotiation auto
 spanning-tree portfast edge

```

Fig. 26. Puerto de Acceso - D1

```

interface GigabitEthernet3/2
 switchport access vlan 8
 switchport mode access
 negotiation auto
 spanning-tree portfast edge
interface GigabitEthernet3/3
 switchport access vlan 13
 switchport mode access
 negotiation auto
 spanning-tree portfast edge

```

Fig. 27. Puerto de Acceso - D2

```

interface GigabitEthernet3/3
 switchport access vlan 8
 switchport mode access
 negotiation auto
 spanning-tree portfast edge

```

Fig. 28. Puerto de Acceso - A1

E. Verify PC to PC connectivity.

Para finalizar la sección de configuraciones en cuanto a conectividad, se procede a verificar el alcance de extremo a extremo, para ello se realizarán pruebas de PING entre los PCs pertenecientes a cada VRF. Es decir, los usuarios de la VLAN 13 (VRF Special Users) tendrán conectividad entre ellos, pero no hacia los usuarios de la VLAN 8 (VRF General Users), de tal forma que se logra concluir que el desarrollo fue adecuado, ya que están totalmente aisladas.

```

PC1 : 10.0.113.50 255.255.255.0 gateway 10.0.113.1

PC1 : 2001:db8:acad:113::50/64

PC1> ping 10.0.213.50

84 bytes from 10.0.213.50 icmp_seq=1 ttl=61 time=24.726 ms
84 bytes from 10.0.213.50 icmp_seq=2 ttl=61 time=11.830 ms
84 bytes from 10.0.213.50 icmp_seq=3 ttl=61 time=13.726 ms
84 bytes from 10.0.213.50 icmp_seq=4 ttl=61 time=15.614 ms
84 bytes from 10.0.213.50 icmp_seq=5 ttl=61 time=15.911 ms

PC1> ping 2001:db8:acad:213::50

2001:db8:acad:213::50 icmp6_seq=1 ttl=58 time=56.739 ms
2001:db8:acad:213::50 icmp6_seq=2 ttl=58 time=15.366 ms
2001:db8:acad:213::50 icmp6_seq=3 ttl=58 time=15.884 ms
2001:db8:acad:213::50 icmp6_seq=4 ttl=58 time=12.013 ms
2001:db8:acad:213::50 icmp6_seq=5 ttl=58 time=10.847 ms

```

Fig. 29. Prueba de conectividad VRF Special Users


```

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  *
*****^C
banner motd ^C R1, ENCOR Skills Assessment, Scenario 2 ^C
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  transport input ssh
line vty 5 15
  transport input ssh
!
no scheduler allocate
!
end

```

B. R2

```

Building configuration...

Current configuration : 5534 bytes
!
version 15.9
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R2
!
boot-start-marker
boot-end-marker
!
!
vrf definition General-Users
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
vrf definition Special-Users
!
  address-family ipv4
  exit-address-family
!
  address-family ipv6
  exit-address-family
!
enable secret 9 $9$1C9D3SJ94AHnu9$0zNC7h9IVRH5caI/RZjNh6gg.4
  BVodT20hMBzIy.JyI
!
aaa new-model
!
!
aaa authentication login default local
!
!
!
!
aaa session-id common
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!

```

```

!
!
no ip icmp rate-limit unreachable
!
!
!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
!
username admin privilege 15 secret 9
$9$TiCsIF2kRRgmWP$SsAwJ8wSXjmfj7cmHBGuB0htWQMkuhkQ3GfSVKGzYoFI
!
!
redundancy
!
no cdp log mismatch duplex
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
  media-type rj45
!
interface GigabitEthernet0/0.1
  encapsulation dot1Q 13
  vrf forwarding Special-Users
  ip address 10.0.12.2 255.255.255.0
  ipv6 address FE80::2:1 link-local
  ipv6 address 2001:DB8:ACAD:12::2/64
!
interface GigabitEthernet0/0.2
  encapsulation dot1Q 8
  vrf forwarding General-Users
  ip address 10.0.12.2 255.255.255.0
  ipv6 address FE80::2:2 link-local
  ipv6 address 2001:DB8:ACAD:12::2/64
!
interface GigabitEthernet0/1
  no ip address
  duplex auto
  speed auto
  media-type rj45
!
interface GigabitEthernet0/1.1
  encapsulation dot1Q 13
  vrf forwarding Special-Users
  ip address 10.0.23.2 255.255.255.0
  ipv6 address FE80::2:3 link-local
  ipv6 address 2001:DB8:ACAD:23::2/64
!
interface GigabitEthernet0/1.2
  encapsulation dot1Q 8
  vrf forwarding General-Users
  ip address 10.0.23.2 255.255.255.0
  ipv6 address FE80::2:4 link-local
  ipv6 address 2001:DB8:ACAD:23::2/64
!
interface GigabitEthernet0/2
  no ip address

```

```

shutdown
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/3
no ip address
shutdown
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/4
no ip address
shutdown
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/5
no ip address
shutdown
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/6
no ip address
shutdown
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/7
no ip address
shutdown
duplex auto
speed auto
media-type rj45
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route vrf General-Users 10.0.108.0 255.255.255.0
GigabitEthernet0/0.2 10.0.12.1
ip route vrf General-Users 10.0.208.0 255.255.255.0
GigabitEthernet0/1.2 10.0.23.3
ip route vrf Special-Users 10.0.113.0 255.255.255.0
GigabitEthernet0/0.1 10.0.12.1
ip route vrf Special-Users 10.0.213.0 255.255.255.0
GigabitEthernet0/1.1 10.0.23.3
!
ipv6 route vrf General-Users 2001:DB8:ACAD:108::/64
GigabitEthernet0/0.2 2001:DB8:ACAD:12::1
ipv6 route vrf Special-Users 2001:DB8:ACAD:113::/64
GigabitEthernet0/0.1 2001:DB8:ACAD:12::1
ipv6 route vrf General-Users 2001:DB8:ACAD:208::/64
GigabitEthernet0/1.2 2001:DB8:ACAD:23::3
ipv6 route vrf Special-Users 2001:DB8:ACAD:213::/64
GigabitEthernet0/1.1 2001:DB8:ACAD:23::3
ipv6 ioam timestamp
!
!
!
!
!
!
control-plane
!
banner exec ^C
*****
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  demonstration and IOS *
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banner incoming ^C
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*
*****^C
banner login ^C
*****
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*
*****^C
banner motd ^C R2, ENCOR Skills Assessment, Scenario 2 ^C
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
transport input ssh
line vty 5 15
transport input ssh
!
no scheduler allocate
!
end

```

C. R3

```

Building configuration...

Current configuration : 5132 bytes
!
version 15.9
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R3
!
boot-start-marker
boot-end-marker
!
!
vrf definition General-Users
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
vrf definition Special-Users
!
address-family ipv4

```


D. D1

```
Current configuration : 4539 bytes
!
! Last configuration change at 19:14:17 UTC Mon Dec 29 2025
!
version 15.2
```

```
interface GigabitEthernet0/3
shutdown
negotiation auto
!
interface GigabitEthernet1/0
```



```

switchport trunk allowed vlan 8,13,99
switchport trunk encapsulation dot1q
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
negotiation auto
channel-group 1 mode desirable
!
interface GigabitEthernet1/1
switchport trunk allowed vlan 8,13,99
switchport trunk encapsulation dot1q
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
negotiation auto
channel-group 1 mode desirable
!
interface GigabitEthernet1/2
shutdown
negotiation auto
!
interface GigabitEthernet1/3
shutdown
negotiation auto
!
interface GigabitEthernet2/0
switchport trunk allowed vlan 8,13,99
switchport trunk encapsulation dot1q
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
negotiation auto
!
interface GigabitEthernet2/1
shutdown
negotiation auto
!
interface GigabitEthernet2/2
shutdown
negotiation auto
!
interface GigabitEthernet2/3
shutdown
negotiation auto
!
interface GigabitEthernet3/0
shutdown
negotiation auto
!
interface GigabitEthernet3/1
shutdown
negotiation auto
!
interface GigabitEthernet3/2
shutdown
negotiation auto
!
interface GigabitEthernet3/3
switchport access vlan 13
switchport mode access
negotiation auto
spanning-tree portfast edge
!
ip forward-protocol nd
!
ip http server
ip http secure-server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr
aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr
aes256-ctr
!
!
!
!
!
!
!
!
!
control-plane
!
banner exec ^C

```

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```
^C
banner incoming ^C
```

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```
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banner login ^C
```

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```
^C
banner motd ^C D1, ENCOR Skills Assessment, Scenario 2 ^C
!
line con 0
  exec-timeout 0 0
  logging synchronous
line aux 0
line vty 0 4
  transport input ssh
line vty 5 15
  transport input ssh
!
!
end

```

E. D2

Building configuration...

```

Current configuration : 4067 bytes
!
! Last configuration change at 19:14:15 UTC Mon Dec 29 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname D2
!
boot-start-marker
boot-end-marker
!

```



```
^C
banner motd ^C D2, ENCOR Skills Assessment, Scenario 2 ^C
!
line con 0
  exec-timeout 0 0
logging synchronous
line aux 0
line vty 0 4
  transport input ssh
line vty 5 15
  transport input ssh
!
!
end
```

```
ip cef
ipv6 unicast-routing
ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
!
!
!
!
!
!
!
!
!
interface Port-channel1
 switchport trunk allowed vlan 8,13,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 switchport nonegotiate
!
interface GigabitEthernet0/0
 shutdown
 negotiation auto
!
interface GigabitEthernet0/1
 shutdown
 negotiation auto
!
interface GigabitEthernet0/2
 shutdown
 negotiation auto
!
interface GigabitEthernet0/3
 shutdown
 negotiation auto
!
interface GigabitEthernet1/0
 switchport trunk allowed vlan 8,13,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 switchport nonegotiate
 negotiation auto
 channel-group 1 mode desirable
!
interface GigabitEthernet1/1
 switchport trunk allowed vlan 8,13,99
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 switchport nonegotiate
 negotiation auto
 channel-group 1 mode desirable
!
interface GigabitEthernet1/2
 shutdown
 negotiation auto
!
interface GigabitEthernet1/3
 shutdown
 negotiation auto
!
interface GigabitEthernet2/0
 shutdown
 negotiation auto
!
interface GigabitEthernet2/1
 shutdown
 negotiation auto
!
interface GigabitEthernet2/2
 shutdown
 negotiation auto
```

F. Al

Building configuration...

```
Current configuration : 4391 bytes
!
! Last configuration change at 19:14:15 UTC Mon Dec 29 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname Al
!
boot-start-marker
boot-end-marker
!
!
enable secret 5 $1$gzlE$XNOHLMIRdRCGuFdjE4wuKl
!
username admin privilege 15 secret 9 $9$vt.WdAlyGw/
ooP$NqfVBQK7.eOUOW7PfXRStOI736jFYlyG7llFOX69MmY
aaa new-model
!
!
aaa authentication login default local
!
!
!
!
!
!
!
!
aaa session-id common
!
!
!
!
!
!
!
!
no ip domain-lookup
```

```

!
interface GigabitEthernet2/3
 shutdown
 negotiation auto
!
interface GigabitEthernet3/0
 shutdown
 negotiation auto
!
interface GigabitEthernet3/1
 shutdown
 negotiation auto
!
interface GigabitEthernet3/2
 shutdown
 negotiation auto
!
interface GigabitEthernet3/3
 switchport access vlan 8
 switchport mode access
 negotiation auto
 spanning-tree portfast edge
!
ip forward-protocol nd
!
ip http server
ip http secure-server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr
 aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr
 aes256-ctr
!
!
!
!
!
!
!
!
control-plane
!
banner exec ^C
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```

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```

^C
banner motd ^C A1, ENCOR Skills Assessment, Scenario 2 ^C
!
line con 0
 exec-timeout 0 0
 logging synchronous
line aux 0
line vty 0 4
 transport input ssh
line vty 5 15
 transport input ssh
!
!
end

```